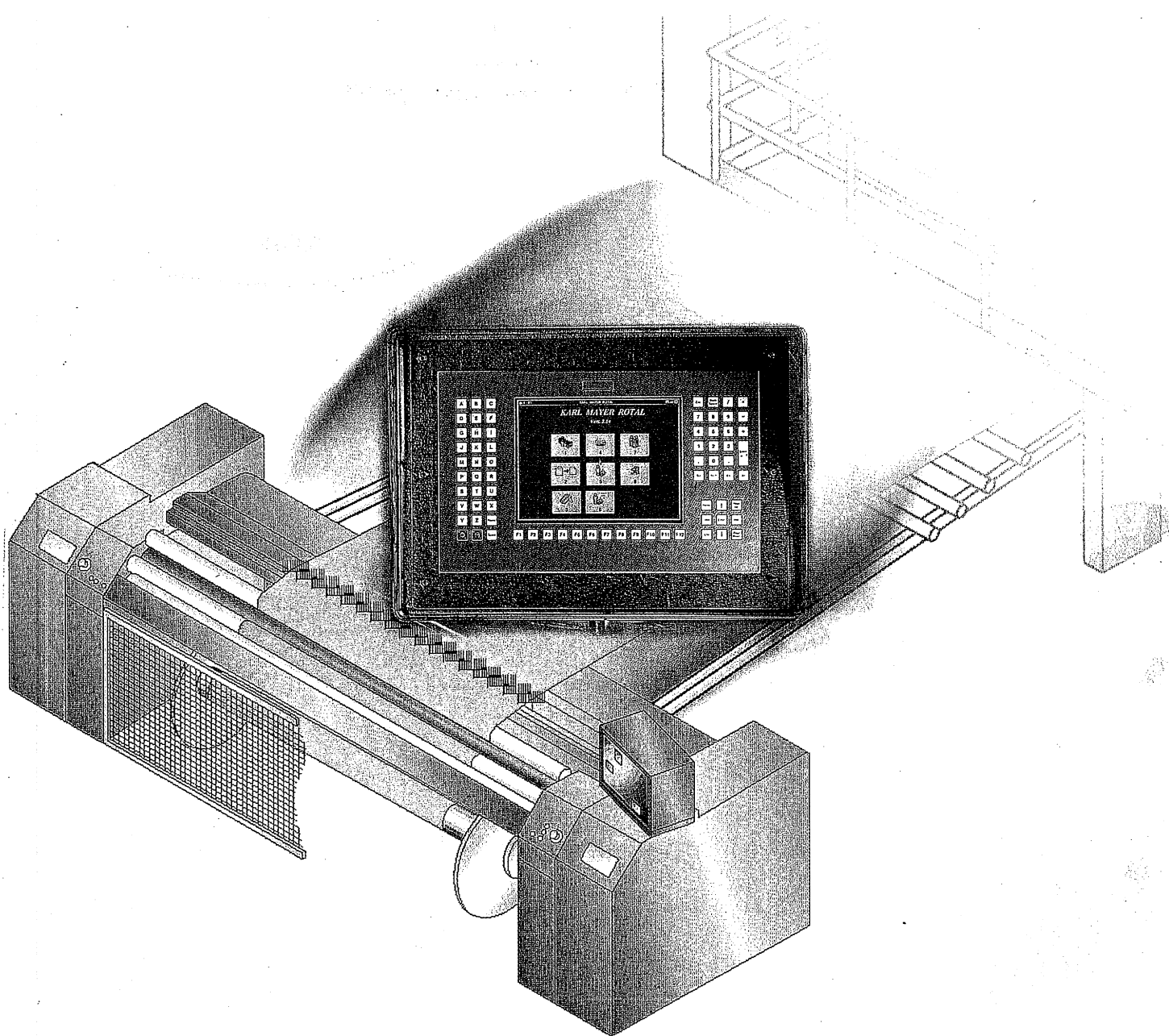




GUIDE TO THE SIZING MACHINE USE

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ROTAL

INTRODUCTION

SIZING

Sizing is a very important step for getting a good output from the loom, and in particular nowadays, as faster and faster machines are used and quality is required at the highest level.

The sizing operation is now carried out on most warps, for the wider use of single yarns, or for yarns in intimate blend with the most different components such as Lycra, polyester, viscose, etc..., to get the most different fabrics with special effects such as stretching or others. Thus, sizing is not only a simple application of a starch product, or anyway a glue (size) to make the fabrics more resistant to pulling and smoother to aid their passage on the weaving machines, stitches, heddles, comb, etc... but it has turned into a true laboratory, where the use of special sizes, the application method and the skill of the operators make the weaving process more and more efficient and profitable.

A saying from the old said "a well prepared and sized warp is already half fabric": indeed, to get a high output and a high weaving quality this is really vital, and thus a good sizing technician must have a well manufactured machine, equipped with every device able to ensure good control levels on tension, elongation, temperature, squeezing, etc..., but he must also be able to choose the necessary parameters according to the items to be processed, their physical-chemical composition and also to choose those that are most suitable for that processing and for the following operations of desizing and dressing.

In the past, and above all when processing clothing wool (drapery and woollens), the so-called size machines were already used; in the Biella province, the best industries were equipped with machines mostly produced in Germany, with copper sizing tanks with squeezing cylinders coated with a cloth. The warps were then moved to chambers with some return cylinders where they were dried with hot air (the most modern machines were equipped with copper drying cylinders heated by steam), then they were crabbed on beams going to the loom.

While processing cotton yarns or blends, sizing has always been the most developed and refined treatment, as twisted yarn, or anyway twofold or multifold yarn, was seldom used, where sizing was not required.

Small companies usually wetted the warp chain with a variable quantity of size using cheap products and relying on the addition of greases or other components to improve the weaving quality.

The new technologies and the checks for pollution and industrial wastes have led to new studies and improvements, both in the sector of products used for sizing and within the new sizing techniques.

Thus, considering the pollution factor (the de-sizing step is considered one of the major causes in the textile process), in the '90s a new processing method was studied, using a n over 15-year experience by a Dutch company; the technique of "wet on wet" has been developing first in Europe and then in the US. The tests carried out in the textile institutes and above all in the ITV in Denkendorf, together with the major manufacturers in this sector, Karl Mayer Rotal, Benninger and Sucker Muller, showed that the breaking load of the warp threads increases by 15 - 19%; it significantly reduces the thread hairiness; it increases the resistance to abrasion with the same sizing; it decreases by 30 - 40% the use of sizing products, increasing the residual elongation.

This technique is nowadays used by the best cotton textile industries, and has been widely disclosed in the sizing conferences held at the ITV of Denkendorf in 1998 and 2001: what the previous studies and tests had indicated was confirmed here. The positive effect of pre-wetting can be explained with 3 reasons:

- 1) some of the hydrophobic substances unrelated to the fiber detach passing through hot water.
- 2) The combined action of the detachment of parts such as pectin, etc..., and of hot water, causing a degumming effect, increases the affinity of the fiber with the size, so that the specific adhesive power increases significantly.
- 3) As the water adhering on the thread fill the gaps between the fibers, the quantity of size bath penetrating the yarn is lower, so that the surface is coated with a more effective and protective coat.

Another significant positive factor of this system is the strong decrease in the consumption of sizing products (about 30-40% less), reducing at the same time the pollution percentage in waste waters.

Obviously, not all yarn types can be processed under this technique; indeed, the shirt garments, or anyway warp chains with several colors cannot be sized with the wet-on-wet system (commonly called Prewet), as the different hydrophilicity of the yarn due to the different dyeing systems or to dyestuff with darker or lighter shades influencing the water permeability level, thus making the sizing degree irregular. The various sizing systems can be configured in a different way according to the type of processing, the type of fiber and the productive capacity; basically 2 systems can be considered, and namely: machines for processing sectional beams (shirt garments with several colour, or wool yarns, or special blended yarns such as wool-poly-Lycra, or for limited production), or machines for fractional beams, where the fractions are loaded on suitable creels and then re-joint and sized for forming the loom beam.

Further, the machines can be equipped with one or more size boxes according to the number of threads or the company needs; the number of drying cylinders determines the machine operating speed, and thus its productivity.

The above description applies exclusively to the processing and thus sizing of animal or vegetable discontinuous fibers; a different technology is instead applied for synthetic or artificial fibers, therefore for yarns composed of several filaments placed reciprocally parallel and thus without any twist; in this case, the systems can be several:

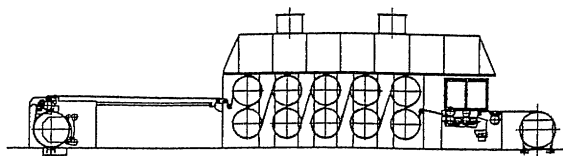
- 1) traditional system: it generally includes a sizing machine that joining at the same time several fractions, after the size soaking it, dries the yarn by contact through the thermal action of drying cylinders, that unlike machines for short fibers must have a temperature control increasing at the start and decreasing in the last cylinders. This type of sizing machine is used only in case of yarn with a min. twist (150 twists/m) and not for twistless yarns; another type of traditional machine uses a pre-drying furnace composed of a chamber where the threads pass, separate both vertically and horizontally. This system is also called "Chemsize" as it was first used by American Chemstrand Corporation; some manufacturers have used radio-frequency furnaces for the pre-drying chambers instead of the standard furnaces with hot water forced recirculation; this system uses an electro-magnetic field generated by the radio-frequency, so that water can be removed from the yarn without heating the threads.
- 2) The second system is called direct sizing and enable to size all the fibers with continuous parallel filament; it is basically different from all the previous ones because the supply of the sizing machine does not take place from the fractions, but as from a fractional warping machine from the thread packages as they are received from their producers. This system is commonly called "Japanese", as it was created first by Tsudakoma.

Summing up, this sizing machine produces a fraction of sized threads, as the warp, while being formed, is soaked with size, dried by the combined action of hot air furnaces and following drying cylinders, then crabbed on beams at an operating speed that can reach 650 m/1'. To originate in the end the true weaving beam the sized fractions are then assembled (joint) on a joining machine that has the beam-holding structure and the beaming head or winder just like sizing machines.

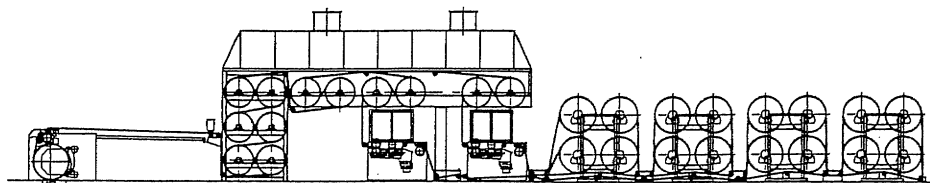
- 3) Sizing from fraction or "beam to beam": it is the same machine described previously in the direct sizing, but instead of being supplied from the original packages (bobbins) it sizes single fractions, previously warped, on special warpers. In this way, even non-perfectly metered bobbins or medium-quality items can be used, as the possible stop due to breakage on the warper does not influence the degree of the following sizing; even in this case, the fractions sized in this way are later joint to originate the weaving beam.

Ending up this overview of the different sizing systems, special attention must be paid to the continuous dyeing-sizing systems for producing warp chains for denim warps, where Italian manufacturers are leader all over the world: they can dye, size and directly prepare the weaving beams in a single passage with following space saving and optimization of the dyeing degree with long meterage. Ending these short remarks on the importance of sizing, it must be mentioned how much is made in this field in Europe and above all in Italy: indeed, the Italian textile industry (due to the strong competition from developing countries with low labour cost) must create newer and newer yarn types with the most different blends, to get more and more refined and special fabrics, or processing very valuable wool with single end, with fineness inconceivable just a few years ago, and that nowadays the designers and industries of our country are instead able to do, making the Italian textile sector and therefore the mechano-textile sector a worldwide pride.

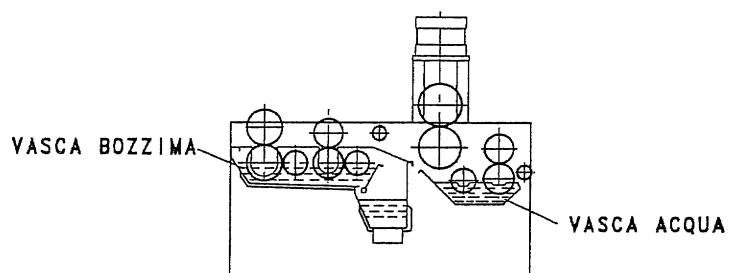
René Drigo



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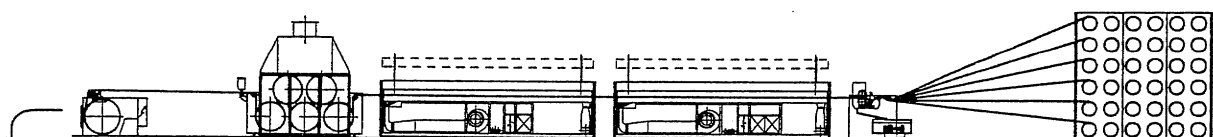


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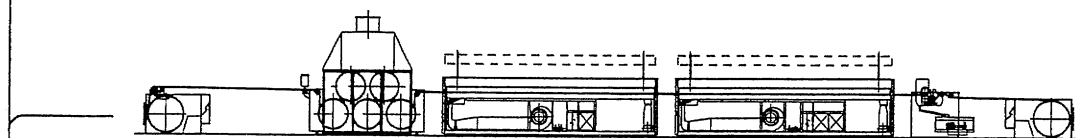


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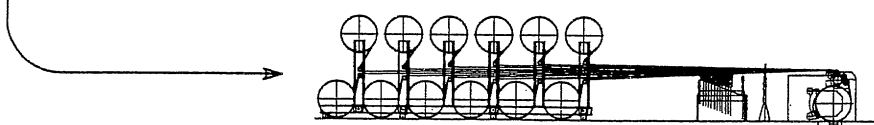
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IMBOZZIMATRICE BEAM TO BEAM



RIUNITRICE

GUIDE FOR USING THE SIZING MACHINE

Sizing was created to give the threads forming the warp the following features:

- sufficient resistance to abrasion
- even elongation and tension
- outer coating to reduce their hairiness as much as possible.

These features must aid a smooth passage of the sized thread along the loom path (warp stop-motion, heddles and comb) to ensure a perfect formation of the step thus achieving a fault-free fabric, or at least the best possible fabric. For a correct sizing the type of used yarn must always be considered, as well as the type of weaving by which it has been produced; in the end the weave used in the loom to obtain the fabric (cloth, twill, sateen...). The choice of the sizing products is also very important, as well as their concentrations and application methods to avoid unwished rheologies in the following processing steps.

For example, if the fabric must be dyed it is indispensable to use completely biodegradable sizes to avoid unpleasant effects of moirè colours or uneven shades. Again, some products cannot be compatible with the following treatments: the use of polyvinyl alcohol at certain concentrations without a previous suitable washing at the singeing could polymerize the PVA residues with crystallizing effects of the products on the fabric, with negative results.

Other example, sizing products - that can be hardly removed at low temperatures, during the dyeing steps in continuous lines with few preliminary washing tanks - can jeopardize the dyeing itself.

Besides the above remarks, it is clear the sizing aids, together with other factors, the weaving output and quality; thus, the sizing operator must know everything concerning the re. yarn: physical features, previous and following treatments and on which machine types to obtain the product final step.

Please, consider that each type of fiber and thread requires different products, machinery, processings and methods.

This little guide will deal and clear the use of a sizing machine for short fibers (cotton, wool, blends...) with drier with water steam heated cylinders, with size boxes with one or two soakings and simple or double squeezing. There are 3 control parameters for the thread physical features:

- 1) elongation
- 2) resistance to abrasion and dynamometric resistance
- 3) hairiness

1) First of all, it is always to be considered that the sized yarn must have a residual elongation equal to 70% of its original elongation; this requirement can be achieved only if all threads show the same tension already while forming the warp beam.

Thus, a good sizing requires a perfect warping, needing in its turn a bobbin winding as careful as possible.

Considering that a perfect finished products can be influenced by yarns coming from different lots, all this indicates how much careful each step of the textile processing must be. Coming back to warping, yarns coming from the same lot should be used, forming bobbins with even tension and the same meterage.

Creels should be equipped with suitable tension units according to the tensions to be exerted on the single threads to guarantee an unwinding and winding on the fractional or sectional beam as much perfect as possible (for cotton, on beams flange 1000, the peripheral difference should not exceed 3 mm); they should also be equipped with more sensitive machine stop systems, as accurate as possible, for a warper fast and accurate intervention to restore any breakage. After these initial remarks, a steady control of the thread tensions in every section of the sizing machine is needed to check the its draft % that can influence directly the size absorption in the size box.

2) The resistance to abrasion and the dynamometric resistance are factors ensuring an efficient treatment on the weaving machines; the choice of the sizing products, together with a more and more perfect control of the processing parameters ensures output standards that are higher and higher and above all reproducible.

Considering the remarkable influence of the products on the warp behaviour in the machine, it is always recommended to discuss or process recipes with the suppliers (above all when processing new items very different from the previous ones).

The positive effect by the use of paraffin treatment or the use of special products, such as antistatics or others, to improve the friction coefficient in the loom yarn guide components should also be considered.

3) The thread outer hairiness is a factor that can be improved and solved with suitable measures and using suitable products; the correct values for squeezing, drying cylinder temperatures and warp tensions in the separation table are thus fundamental. It can be easily understood that a sudden opening of the various thread layers, above all if still too wet, besides breaking the outer film of the applied size, also significantly increases the presence of surface fibers not wound in the thread body, that while weaving could originate the defect of inwound dust, increase the warp breakage and thus decrease the loom output, increase the dustiness in the loom rooms.

As for the squeezing, the right compromise must be searched for each warp type between size penetration inside the thread and outer coverage of the thread itself. The thread twist coefficient is thus extremely important; a yarn oe should usually have a @ of 115-150, while a ring yarn 100-110 (cotton base).

Before considering the setting of the different working parameters it is advisable to consider a few fundamental basis for a perfect warp preparation. It is thus recommended:

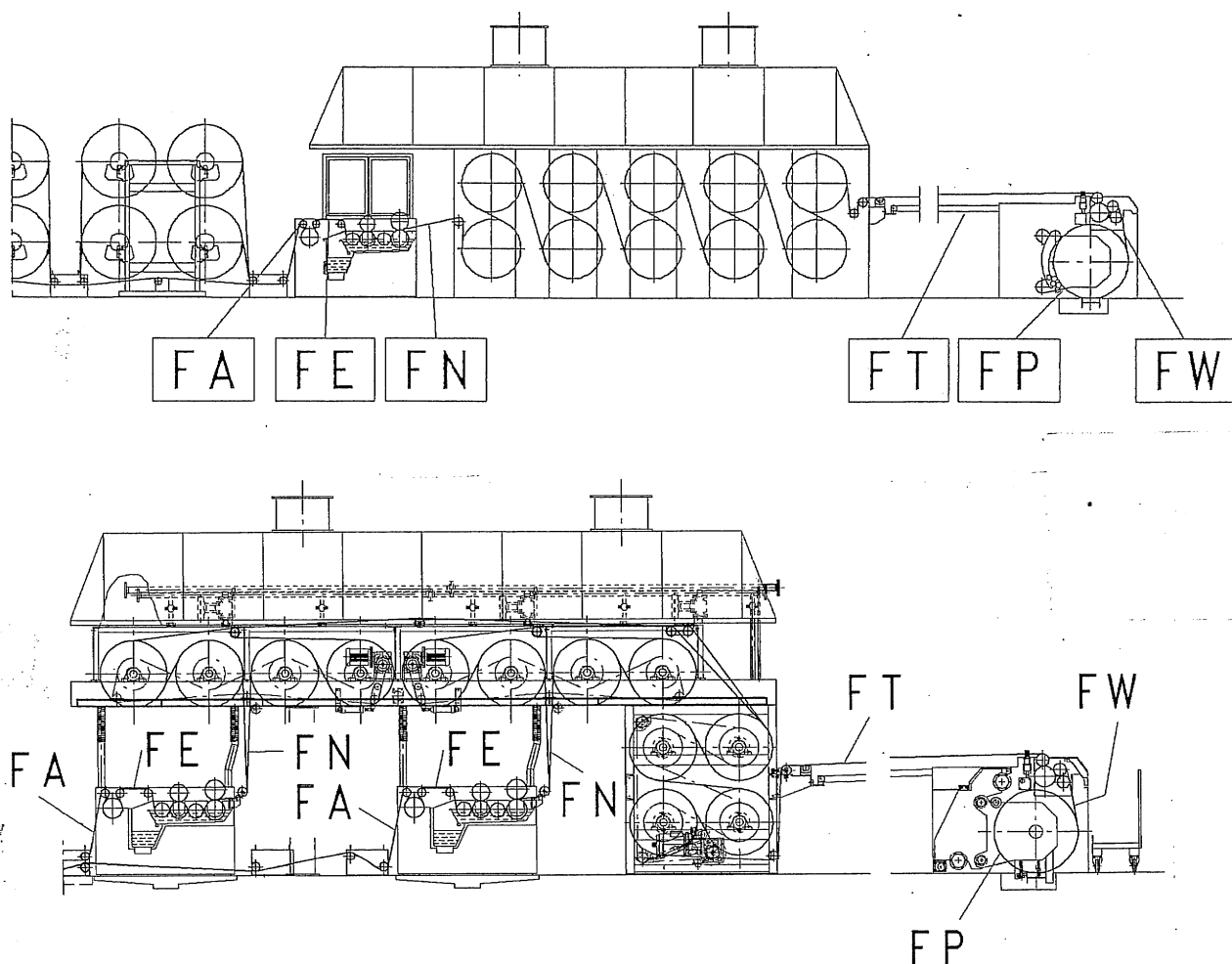
- to check the alignment of the fractions on the creel;
- to unwind the warp beams carefully, possibly fixing the layer of threads of each single fraction with some adhesive tape to prevent thread overlaying or crossing;
- the sequence of the various beams must be made according to the required treatment type, if sizing with 1 or 2 size boxes according to the number of threads, always considering that a high density of threads/cm over 85% jeopardizes a good sizing. This is why it is advisable to separate the warp for the passage in 2 size boxes.

Example of calculation for adjusting the cotton and staple tensions

Description	Chain weight (g/m)	Cotton (tension %)	Staple	
			dry (tension %)	wet (tension %)
Unwinding tension FA		2.5–5	2.5–5	2.5–5
Introduction tension FE		1–2.5	1–2.5	1–1.5
Drying entry tension FN		2–3	2–3	2–3
Dividing table tension FT	30–100	8–11	7–9	6–8
	100–150	6.5–8.5	6–7	5–6
	150–300	6–8	4–6	3.5–5
Winding tension FW	30–100	10–14	9–12	8–10
	100–150	9–12	6–9	5–8
	150–300	8–10	5–7	4–5
Pressure on beam FP		Factor A	Factor A	Factor A
	30–100	2–2.5	1.3–1.5	1.3–1.5
	100–150	1.5–2	0.9–1.3	0.9–1.3
	150–300	1–1.5	0.7–0.9	0.7–0.9

Breaking resistance of cotton yarn 12 km = 12 cN/tex

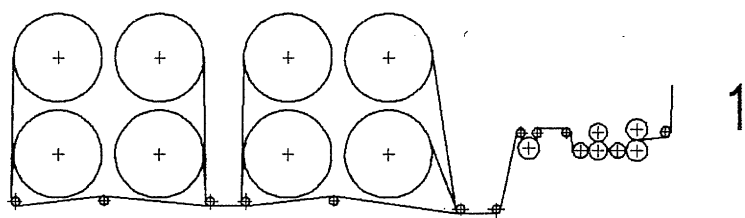
Breaking resistance of staple yarn 14 km = 14 cN/tex



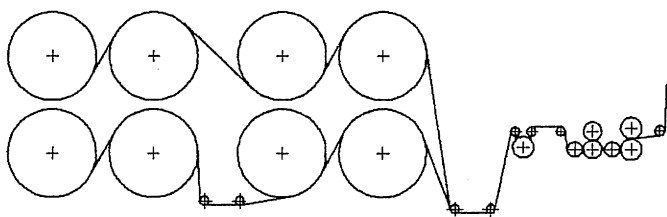
Coming back to the arrangement of the warping beams there are several possibilities, all correct and functional, even if some ensure an even easier intervention for repairing the broken or crossed threads, and some other are more suitable for the continuous dyeing - sizing lines.

Let's see the most used ones:

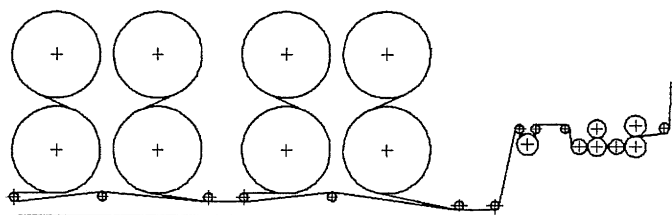
- 1) this is the most used solution, enabling an even unwinding of the warp beams
- 2) this is the solution used most frequently in the continuous systems, where a safe splitting is strictly necessary
- 3) this unwinding ensures a perfect intervention possibility, giving the operator the possibility of restoring the breakage without any interference among beams
- 4) this is the ideal solution for linen yarns, both with dry and wet process.



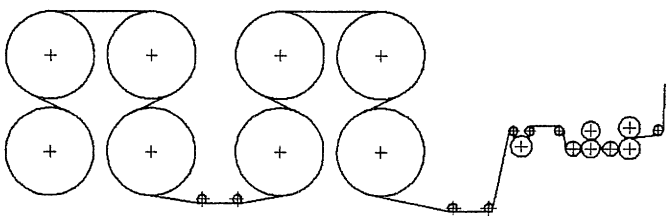
1



2



3



4

Many other solutions can be adopted, but it is anyway advisable to not to make the beams rotate in the same direction, so as to avoid possible warping faults to multiply while sizing; another important factor is the easy availability of separation or classifying bands (usually, thin bands of twisted yarn in polyester or nylon are used, even better a flat braided cord).

It is also advisable to separate the 2 halves practically to aid the selection in the dry splitting range. While sizing, the threads could break and wind on the size box rollers, above all on the squeezer lower rollers.

In this case the operator will notice a sudden separation of the warp chain on the drying cylinder, clearly indicating a breakage with relevant thread winding. First of all, the operator must separate the layer of threads entering the size box, so as to insulate on the side the thread ring formed on the cylinder.

Now the machine can be stopped and it is possible to intervene cautiously: have the system started up with the aid of another operator using the installed man push button.

To remove wound thread, it is advisable to use the suitable curved knife taking care not to etch nor to dent the squeezing cylinders and the teflon-coated cylinders; thus, the blade must always be laid flat making the knife slide on the surface and thus cutting the thread.

Special care must be paid to introducing the suitable separation wires: they must be introduced while starting between the fraction layers, starting from the center line (the one dividing the upper and lower layers).

Be careful and avoid some threads to deviate from their position and to cross reciprocally.

The comb rising is also very important, as it must occur only when the layers are well separate and stretched and the threads are well parallel.

This operation can be made following different methods according to the warp coming from a sectional or fractional warper.

In the first case some operators use a comb with flat teeth and positioning it at the exit of the drying cylinders, immediately after the passage of the so-called cross-piece, they take it up to the comb of the head and lift it.

In the second case, instead, a valid system is positioning some adhesive tape on the cylinder before the waxing device after all the classifying and adhesive tapes applied to each single layer have passed; take it up to the comb taking care not to disarrange the thread carpet.

Another system, always for chains coming from fractional warpers, is dividing the threads on the comb starting from the center and, after a suitable calculation for an exact splitting, then arranging them in suitable number on each teeth passage.

After this operation (both introducing the wires and separating the comb) the warp length is adapted according to the height of the loom beam; obviously, also the choice of the pressing rollers must adapt to the loom beams.

A parameter for a machine correct operation, but above all to ensure an excellent thread treatment both while sizing and later while weaving, is the preliminary setting of the tensions in the various machine points according to the item being processed. Considering the total number and the count of the warp threads, first of all the weight in g/m and the coverage degree are calculated. With these 2 parameters it is possible (by using the enclosed tables) to set the machine operating values and to understand if the type of machine available is the most suitable one for the required treatment.

The decision of using 1 or 2 size boxes depends on the coverage degree (threads/cm * working table); for size boxes with double traditional squeezing this value must not exceed 85%; with the new prewetting systems (our type MPD - PW) it must be equal to 105%.

If the value of the coverage degree is higher, it is necessary to split the warp in 2 layers and then operate on 2 size boxes.

CALCULATION OF THE WARP WEIGHT

L = length of the warp in meters

With metric count $\rightarrow (L * \text{thread no.}) / \text{Nm} = \text{grams} / \text{m}$

With count in deniers $\rightarrow (L * \text{thread no.} * \text{den}) / 9000 = \text{grams} / \text{m}$

With count in decitex $\rightarrow (L * \text{thread no.} * \text{dtex}) / 10,000 = \text{grams} / \text{m}$

CALCULATION OF THE COVERAGE

As said before, this data item is extremely important for a correct process; indeed, an excessive coverage hinders an even sizing and thus may require a warp splitting and the process on 2 size boxes. On the other hand, a low coverage ($< \text{ or } = 40\%$) could cause twisting and cord forming with some yarns.

To decide if the coverage is correct or if the warp should be split, refer to the theoretical value indicating how many threads can be arranged one next to the other in 1 cm. This value is considered as parameter of the density equal to 100%.

The enclosed table shows this value in the last column (C 100 ENDS / CM).

Please note that for calculating these values, 1 dry thread with density 0.8 g/cm^3 has been considered.

The actual coverage is calculated in % according to the following method:

$$\text{actual coverage \%} = (\text{thread tot. no.} / \text{useful height in cm}) / \text{C 100}$$

Example:

let's consider an item with total 9000 threads

useful height cm 180

thread count Nm 85

$$9000 / 180 = 50 \text{ threads} / \text{cm}$$

The comparison table shows that for a yarn Nm 85 the value CI 00 corresponds to a coverage of 100% equal to 73.0%.

Thus, the effective coverage for this item will be $(50 / 73) * 100 = 68.5$

As for the calculation of the tension, refer to the enclosed pages 2.4.1. - 2.4.2 and example). For a fast machine start-up some tables have been prepared, for an easy reference based on the weight in g/m of the warp; indeed, starting from this value the tension parameters to be set in the various machine points are determined, considering that they are indicative and thus could change by $\pm 15\%$ according to the type of thread (tables A - B - C).

For a perfect treatment of the warps on the weaving machines, besides the evenness of the sizing, the residual humidity of the yarn must also be carefully controlled, obviously changing according to the various fiber types.

Data:	cotton yarns	5%	-	7 %
	wool yarns	6.5%	-	9 %
	acetate	7%	-	9 %
	viscose - cupro	8%	-	10 %
	cellulose	5%	-	6%

The current machine usually have roller control devices (feelers) that, depending on the set humidity, automatically adjust the operating speed to keep the set value steady.
 Besides this, a suitable table placed on the instrument indicates the water comparison values for pure fibers (see table D).

SIZING PRODUCTS

STARCHES AND POWDERS

Historically, starches and powders have been the base for textile thread sizing; nowadays, powders are less used than in the past, while starches are still extremely important, above all thanks to their derivatives that have significant advantages, even if higher costs. Among these, let's mention oxidized, etherized, esterized and carboxymethylated starches. These products offer the following advantages:

- easy and fast preparation
- lower and steady viscosity
- stability and fluidity even at low temperatures

CMC (CarboxyMethylCellulose)

Type of products that, after the category of starches, is remarkably important for the yarn sizing.

Those available on the market differ in their pureness degree (pure cmc - technical cmc) and are classified according to their viscosity in low, medium, medium/high and high.

FORMULATIONS OR COMPOUNDS

They are mixtures of polymers and additives suitably studied to prevent the user from wasting time, and are divided in 2 categories: reinforcing and ready to use.

The first ones correct a recipe already studied by the sizing operator and are usually adopted in small quantities, taking the recipe viscosity to the wished values to ensure a correct thread coverage.

The second ones contain all the ingredients needed for processing certain yarns containing additives such as softeners, antistatic products, as well as PVA, CMS, CMC...

SIZES BASED ON SYNTHETIC POLYMERS

These products are used for sizing the warp continuous threads and significantly aid also the treatment of discontinuous yarns (cotton, wool, viscose and their blends) mixed with products such as modified starches, carboxymethylates, guaro, etc...

Synthetic polymers can be roughly divided into mPolyVinyl Alcohol (PVA) - acrylic copolymers - vinyl copolymers - styrene copolymers - water-soluble or water-dispersible polyurethane polyesters; obviously, combinations of the families of the above mentioned polymers can be obtained with special procedures, for example polyvinylacrylates and acrylate polyesters, etc...

To sum up and avoid treating special products too long, we will deal with what is actually considered the size par excellence, namely PVA.

Marketed since 1939, it is a synthetic resin and the products available on the market are generally featured by a number indicating their hydrolysis degree, ranging from 80% to 99%, and by viscosity that, like for cmc, is divided in low, medium, medium / high, high. The main feature of PVAs is their high solubility, featured by the polymerization degree. They can be easily removed by washing at high temperature and have an outstanding tensile strength compared to other sizing products.

Just consider that a film made up of maize starch has a resistance of 700 - 800 psi, whereas for a cmc the value reaches 7000 psi and for PVA the value rises to 22000 psi.

Another very important point for the environmental impact and above all for water pollution is the low BOD (biological oxygen demand): a generic starch has a value of 477 ppm of a 0.1% solution, PVA has a BOD of 16 ppm, equal to the one of a solution with the same cmc concentration. A further positive feature of PVA is that, with the same resistance of the sizing films, the usage quantity is definitely lower compared to the one of starch-based products. Polyvinyl alcohols absorb and release humidity to balance the room relative humidity; the balance of humidity content of the PVA film depends on the crystallizing and orientation degree of the molecules, and thus on the hydrolysis degree. The modern textile technology requires increasingly demanding performance: what was considered an almost unsurpassable limit as a weft introduction of 1500 m/min is nowadays widely surpassed.

The warp preparation is thus more and more refined and studied according to the company needs. PVA, too, is going to be gradually replaced by its derivatives, the modern acrylic or vinyl resins that, suitably selected for the different yarn types, guarantee an excellent result, above all for oe yarns that, due to their nature, tend to absorb high size quantities; with high viscosity products at low usage concentrations (5% - 6%) a perfect outer coverage can be achieved (elastic protective coating, resistant to abrasion). The same result is achieved also for sizing linen, where the use of a purified cmc with high viscosity enables to work with concentrations of 4% - 5%. In the shirt sector, the use of PVA derivatives with cmc mixtures and the addition of soaking and smoothing agents (compounds) enable processing with looms at 600 rpm, with high output and quality.

A special remark must be made for the use of the modern working method called PRE- WET, namely prewetting; this system enables (on raw yarns) to work with a coverage of 100% and over, to increase by 1-2% the output of the weaving room, to obtain significant savings on sizing products and energy, as well as on waste water purification costs.

Obviously, the sizing technician must consider (when formulating the sizing recipe) the humidity contribution due to the soaking in hot water, thus he shall increase the recipe by about 1.5-2% to balance the decrease of the concentration in the size box, because of the thread water contribution. Generally, the yarn is calculated to acquire 30-35% of humidity and therefore there will be a load of dry sizing agent of 65-70%, that being a film applied outside the thread will get a higher residual elongation.

Nm	Ne	tex	den	C100 (ends/cm)
1	0,59	1000,00	9000	7,9
2	1,18	500,00	4500,00	11,2
3	1,77	333,33	3000,00	13,7
4	2,36	250,00	2250,00	15,8
5	2,95	200,00	1800,00	17,7
6	3,54	166,67	1500,00	19,4
7	4,13	142,86	1285,71	20,9
8	4,72	125,00	1125,00	22,4
9	5,31	111,11	1000,00	23,7
10	5,90	100,00	900,00	25,0
11	6,49	90,91	818,18	26,3
12	7,08	83,33	750,00	27,4
13	7,67	76,92	692,31	28,5
14	8,26	71,43	642,86	29,6
15	8,85	66,67	600,00	30,7
16	9,44	62,50	562,50	31,7
17	10,03	58,82	529,41	32,6
18	10,62	55,56	500,00	33,6
19	11,21	52,63	473,68	34,5
20	11,80	50,00	450,00	35,4
21	12,39	47,62	428,57	36,3
22	12,98	45,45	409,09	37,1
23	13,57	43,48	391,30	38,0
24	14,16	41,67	375,00	38,8
25	14,75	40,00	360,00	39,6
26	15,34	38,46	346,15	40,4
27	15,93	37,04	333,33	41,1
28	16,52	35,71	321,43	41,9
29	17,11	34,48	310,34	42,6
30	17,70	33,33	300,00	43,4
31	18,29	32,26	290,32	44,1
32	18,88	31,25	281,25	44,8
33	19,47	30,30	272,73	45,5
34	20,06	29,41	264,71	46,2
35	20,65	28,57	257,14	46,8
36	21,24	27,78	250,00	47,5
37	21,83	27,03	243,24	48,1
38	22,42	26,32	236,84	48,8
39	23,01	25,64	230,77	49,4
40	23,60	25,00	225,00	50,1
41	24,19	24,39	219,51	50,7
42	24,78	23,81	214,29	51,3
43	25,37	23,26	209,30	51,9
44	25,96	22,73	204,55	52,5
45	26,55	22,22	200,00	53,1
46	27,14	21,74	195,65	53,7
47	27,73	21,28	191,49	54,3
48	28,32	20,83	187,50	54,8

Nm	Ne	tex	den	C100 (ends/cm)
49	28,91	20,41	183,67	55,4
50	29,50	20,00	180,00	56,0
51	30,09	19,61	176,47	56,5
52	30,68	19,23	173,08	57,1
53	31,27	18,87	169,81	57,6
54	31,86	18,52	166,67	58,2
55	32,45	18,18	163,64	58,7
56	33,04	17,86	160,71	59,2
57	33,63	17,54	157,89	59,8
58	34,22	17,24	155,17	60,3
59	34,81	16,95	152,54	60,8
60	35,40	16,67	150,00	61,3
61	35,99	16,39	147,54	61,8
62	36,58	16,13	145,16	62,3
63	37,17	15,87	142,86	62,8
64	37,76	15,63	140,63	63,3
65	38,35	15,38	138,46	63,8
66	38,94	15,15	136,36	64,3
67	39,53	14,93	134,33	64,8
68	40,12	14,71	132,35	65,3
69	40,71	14,49	130,43	65,8
70	41,30	14,29	128,57	66,2
71	41,89	14,08	126,76	66,7
72	42,48	13,89	125,00	67,2
73	43,07	13,70	123,29	67,6
74	43,66	13,51	121,62	68,1
75	44,25	13,33	120,00	68,6
76	44,84	13,16	118,42	69,0
77	45,43	12,99	116,88	69,5
78	46,02	12,82	115,38	69,9
79	46,61	12,66	113,92	70,4
80	47,20	12,50	112,50	70,8
81	47,79	12,35	111,11	71,2
82	48,38	12,20	109,76	71,7
83	48,97	12,05	108,43	72,1
84	49,56	11,90	107,14	72,5
85	50,15	11,76	105,88	73,0
86	50,74	11,63	104,65	73,4
87	51,33	11,49	103,45	73,8
88	51,92	11,36	102,27	74,3
89	52,51	11,24	101,12	74,7
90	53,10	11,11	100,00	75,1
91	53,69	10,99	98,90	75,5
92	54,28	10,87	97,83	75,9
93	54,87	10,75	96,77	76,3
94	55,46	10,64	95,74	76,7
95	56,05	10,53	94,74	77,2
96	56,64	10,42	93,75	77,6

Nm	Ne	tex	den	C100 (ends/cm)
97	57,23	10,31	92,78	78,0
98	57,82	10,20	91,84	78,4
99	58,41	10,10	90,91	78,8
100	59,00	10,00	90,00	79,2
101	59,59	9,90	89,11	79,6
102	60,18	9,80	88,24	79,9
103	60,77	9,71	87,38	80,3
104	61,36	9,62	86,54	80,7
105	61,95	9,52	85,71	81,1
106	62,54	9,43	84,91	81,5
107	63,13	9,35	84,11	81,9
108	63,72	9,26	83,33	82,3
109	64,31	9,17	82,57	82,6
110	64,90	9,09	81,82	83,0
111	65,49	9,01	81,08	83,4
112	66,08	8,93	80,36	83,8
113	66,67	8,85	79,65	84,1
114	67,26	8,77	78,95	84,5
115	67,85	8,70	78,26	84,9
116	68,44	8,62	77,59	85,3
117	69,03	8,55	76,92	85,6
118	69,62	8,47	76,27	86,0
119	70,21	8,40	75,63	86,3
120	70,80	8,33	75,00	86,7
121	71,39	8,26	74,38	87,1
122	71,98	8,20	73,77	87,4
123	72,57	8,13	73,17	87,8
124	73,16	8,06	72,58	88,1
125	73,75	8,00	72,00	88,5
126	74,34	7,94	71,43	88,9
127	74,93	7,87	70,87	89,2
128	75,52	7,81	70,31	89,6
129	76,11	7,75	69,77	89,9
130	76,70	7,69	69,23	90,3
131	77,29	7,63	68,70	90,6
132	77,88	7,58	68,18	90,9
133	78,47	7,52	67,67	91,3
134	79,06	7,46	67,16	91,6
135	79,65	7,41	66,67	92,0
136	80,24	7,35	66,18	92,3
137	80,83	7,30	65,69	92,7
138	81,42	7,25	65,22	93,0
139	82,01	7,19	64,75	93,3
140	82,60	7,14	64,29	93,7
141	83,19	7,09	63,83	94,0
142	83,78	7,04	63,38	94,3
143	84,37	6,99	62,94	94,7
144	84,96	6,94	62,50	95,0

Nm	Ne	tex	den	C100 (ends/cm)
145	85,55	6,90	62,07	95,3
146	86,14	6,85	61,64	95,6
147	86,73	6,80	61,22	96,0
148	87,32	6,76	60,81	96,3
149	87,91	6,71	60,40	96,6
150	88,50	6,67	60,00	96,9
151	89,09	6,62	59,60	97,3
152	89,68	6,58	59,21	97,6
153	90,27	6,54	58,82	97,9
154	90,86	6,49	58,44	98,2
155	91,45	6,45	58,06	98,5
156	92,04	6,41	57,69	98,9
157	92,63	6,37	57,32	99,2
158	93,22	6,33	56,96	99,5
159	93,81	6,29	56,60	99,8
160	94,40	6,25	56,25	100,1
161	94,99	6,21	55,90	100,4
162	95,58	6,17	55,56	100,8
163	96,17	6,13	55,21	101,1
164	96,76	6,10	54,88	101,4
165	97,35	6,06	54,55	101,7
166	97,94	6,02	54,22	102,0
167	98,53	5,99	53,89	102,3
168	99,12	5,95	53,57	102,6
169	99,71	5,92	53,25	102,9
170	100,30	5,88	52,94	103,2
171	100,89	5,85	52,63	103,5
172	101,48	5,81	52,33	103,8
173	102,07	5,78	52,02	104,1
174	102,66	5,75	51,72	104,4
175	103,25	5,71	51,43	104,7
176	103,84	5,68	51,14	105,0
177	104,43	5,65	50,85	105,3
178	105,02	5,62	50,56	105,6
179	105,61	5,59	50,28	105,9
180	106,20	5,56	50,00	106,2
181	106,79	5,52	49,72	106,5
182	107,38	5,49	49,45	106,8
183	107,97	5,46	49,18	107,1
184	108,56	5,43	48,91	107,4
185	109,15	5,41	48,65	107,7
186	109,74	5,38	48,39	108,0
187	110,33	5,35	48,13	108,2
188	110,92	5,32	47,87	108,5
189	111,51	5,29	47,62	108,8
190	112,10	5,26	47,37	109,1

TABLE OF THE TENSIONS FOR COTTON YARNS (A)

Warp weight	FA	FE	FN	FT	FW	FP
(gr / mt)	(N)	(N)	(N)	(N)	(N)	(N)
40	380	180	250	740	798	1060
50	410	200	270	800	892	1150
60	430	220	290	870	987	1240
70	450	240	320	930	1081	1340
80	470	260	340	990	1176	1430
90	500	280	370	1060	1271	1520
100	520	300	390	1120	1365	1610
110	540	320	420	1190	1460	1710
120	570	340	440	1250	1554	1800
130	590	360	470	1310	1649	1890
140	610	380	490	1380	1733	1980
150	630	400	510	1440	1827	2080
160	660	420	540	1500	1922	2170
170	680	440	560	1570	2016	2260
180	700	460	590	1630	2111	2350
190	730	480	610	1690	2205	2440
200	750	500	640	1760	2300	2540
210	770	520	660	1820	2394	2630
220	790	540	690	1880	2489	2720
230	820	560	710	1950	2583	2810
240	840	580	730	2010	2678	2910
250	860	600	760	2070	2772	3000
260	890	620	780	2140	2867	3090
270	910	640	810	2200	2961	3180
280	930	660	830	2260	3045	3280
290	950	680	860	2330	3140	3370
300	980	700	880	2390	3234	3460
310	1000	720	910	2450	3329	3550
320	1020	740	930	2520	3423	3650
330	1050	760	950	2580	3518	3740
340	1070	780	980	2640	3612	3830
350	1090	800	1000	2710	3707	3920
360	1110	820	1030	2770	3801	4020
370	1140	840	1050	2840	3896	4110
380	1160	860	1080	2900	3990	4200
390	1180	880	1100	2960	4085	4290
400	1210	900	1130	3030	4179	4390
410	1230	920	1150	3090	4274	4480
420	1250	940	1170	3150	4368	4570
430	1270	960	1200	3220	4452	4560
440	1300	980	1220	3280	4547	4760
450	1320	1000	1250	3340	4641	4850
460	1340	1020	1270	3410	4736	4940
470	1370	1040	1300	3470	4830	5030
480	1390	1060	1320	3530	4925	5130
490	1410	1080	1350	3600	5019	5220
500	1430	1100	1370	3660	5114	5310
510	1460	1120	1390	3720	5208	5400
520	1480	1140	1420	3790	5303	5490
530	1500	1160	1440	3850	5397	5590
540	1530	1180	1470	3910	5492	5680
550	1550	1200	1490	3980	5586	5770
560	1570	1220	1520	4040	5681	5860
580	1615	1260	1565	4165	5859	6050
600	1660	1300	1610	4300	6048	6230

A seconda del tipo di filato le tensioni indicate potranno variare fino a $\pm 15\%$

TABLE OF THE TENSIONS FOR POLYESTER-COTTON YARNS (B)

Warp weight (gr / mt)	FA (N)	FE (N)	FN (N)	FT (N)	FW (N)	FP (N)
40	440	250	320	740	1000	1220
50	470	270	350	810	1080	1330
60	490	290	380	880	1160	1430
70	520	320	410	950	1250	1540
80	550	340	440	1020	1330	1640
90	580	360	460	1100	1410	1750
100	600	390	490	1170	1490	1850
110	630	410	520	1240	1580	1960
120	660	440	550	1310	1660	2060
130	690	460	580	1380	1740	2170
140	710	480	610	1460	1830	2270
150	740	510	640	1530	1910	2380
160	770	530	670	1600	1990	2480
170	790	560	690	1670	2080	2590
180	820	580	720	1740	2160	2690
190	850	600	750	1810	2240	2800
200	880	630	780	1890	2330	2900
210	900	650	810	1960	2410	3010
220	930	680	840	2030	2490	3120
230	960	700	870	2100	2570	3220
240	990	720	900	2170	2660	3330
250	1010	760	930	2250	2740	3430
260	1040	770	950	2320	2820	3540
270	1070	790	980	2390	2910	3640
280	1090	820	1010	2460	2990	3750
290	1120	840	1040	2530	3070	3850
300	1150	870	1070	2610	3160	3960
310	1180	890	1100	2680	3240	4060
320	1200	910	1130	2750	3320	4170
330	1230	940	1160	2820	3410	4270
340	1260	960	1180	2890	3490	4380
350	1290	990	1210	2970	3570	4480
360	1310	1010	1240	3040	3650	4590
370	1340	1030	1270	3110	3740	4690
380	1370	1060	1300	3180	3820	4800
390	1390	1080	1330	3250	3900	4910
400	1420	1100	1360	3320	3990	5010
410	1450	1130	1390	3400	4070	5120
420	1460	1150	1420	3470	4150	5220
430	1500	1180	1440	3540	4240	5330
440	1530	1200	1470	3610	4320	5430
450	1560	1220	1500	3680	4400	5540
460	1590	1250	1530	3760	4490	5640
470	1610	1270	1560	3830	4570	5750
480	1640	1300	1590	3900	4650	5850
490	1670	1320	1620	3970	4730	5960
500	1690	1340	1650	4040	4820	6060
510	1720	1370	1670	4120	4900	6170
520	1750	1390	1700	4190	4980	6270
530	1780	1420	1730	4260	5070	6380
540	1800	1440	1760	4330	5150	6480
550	1830	1460	1790	4400	5230	6590
560	1860	1490	1820	4470	5320	6690
580	1910	1530	1880	4620	5480	6910
600	1970	1580	1930	4760	5650	7120

A seconda del tipo di filato le tensioni indicate potranno variare fino a + / - 15 %

TABLE OF THE TENSIONS FOR STAPLE YARNS (C)

Warp weight (gr / mt)	FA (N)	FE (N)	FN (N)	FT (N)	FW (N)	FP (N)
40	240	120	180	560	680	800
50	270	140	210	620	760	860
60	300	160	230	680	830	910
70	330	180	260	740	900	970
80	360	200	290	810	980	1030
90	390	220	310	870	1050	1090
100	420	240	340	930	1120	1150
110	450	260	370	990	1200	1210
120	490	290	390	1050	1270	1270
130	520	310	420	1120	1340	1330
140	550	330	450	1180	1420	1390
150	580	350	470	1240	1490	1450
160	610	370	500	1300	1560	1510
170	640	390	530	1360	1640	1560
180	670	410	550	1420	1710	1620
190	710	440	580	1490	1780	1680
200	740	460	610	1550	1860	1740
210	770	480	640	1610	1930	1800
220	800	500	660	1670	2000	1860
230	830	520	690	1730	2080	1920
240	860	540	720	1800	2150	1980
250	890	560	740	1860	2220	2040
260	929	590	770	1920	2300	2100
270	960	610	800	1980	2370	2160
280	990	630	820	2040	2440	2210
290	1020	650	850	2100	2520	2270
300	1050	670	880	2170	2590	2330
310	1080	690	900	2230	2660	2390
320	1110	710	930	2290	2740	2450
330	1140	740	960	2350	2810	2510
340	1180	760	980	2410	2880	2570
350	1210	780	1010	2480	2960	2630
360	1240	800	1040	2540	3030	2690
370	1270	820	1060	2600	3100	2750
380	1300	840	1090	2660	3180	2810
390	1330	860	1120	2720	3250	2860
400	1360	890	1140	2780	3320	2920
410	1400	910	1170	2850	3400	2980
420	1430	930	1200	2910	3470	3040
430	1460	950	1230	2970	3540	3100
440	1490	970	1250	3030	3620	3160
450	1520	990	1280	3090	3690	3220
460	1550	1010	1310	3160	3760	3280
470	1580	1040	1330	3220	3840	3340
480	1610	1060	1360	3280	3910	3400
490	1650	1080	1390	3340	3980	3460
500	1680	1100	1410	3400	4060	3510
510	1710	1120	1440	3460	4130	3570
520	1740	1140	1470	3530	4200	3630
530	1770	1160	1490	3590	4280	3690
540	1800	1190	1520	3650	4350	3750
550	1830	1210	1550	3710	4420	3810
560	1870	1230	1570	3770	4500	3870
580	1930	1270	1630	3900	4640	3990
600	1990	1310	1680	4020	4790	4110

A seconda del tipo di filato le tensioni indicate potranno variare fino a + / - 15 %

TABLE OF THE H2O COMPARISON VALUES FOR PURE FIBERS

Cotton	1,5	2,5	3	4	5	6	7	8	9	10	11	12
Mer. cot.	4	4,8	5,3	6,5	7,5	8,5	9,9	10,9	11,9	12,4	13,9	14,9
Silk	2,8	5,1	6	8,1	10,1	11,6	12,5	13,3	14,3	14,8	16,5	18
Acrylic	0,5	0,9	1	1,4	1,8	2,1	2,5	2,8	3,2	3,3	3,8	4,2
Viscose	3,5	4,1	5,2	6,8	9,6	11,1	12,4	13,9	15,2	15,9	18,5	19,8
Wool	8,5	9,5	13,5	14,5	16,5	18	21	22	23,5	24,5	25,5	26
Polyester	0,25	0,37	0,45	0,6	0,75	0,9	1,06	1,2	1,35	1,42	1,65	1,8
Acetate	2,5	3	3,6	4,3	5,4	6,5	7,6	8,7	9,8	10,3	11,6	12,4
Nylon	2	2,5	3,4	4,6	5,7	6,8	8,1	9,1	10,3	10,8	12,3	13,7
Linen	3	4	5	6	7	8	9	10	10,5	11	11,5	13
Cellulose	3	4,1	4,5	5,7	6,8	7,8	8,8	9,9	10,9	11,5	13	14
70W. - 30A.	1,8	2	2,8	3,4	5	6,6	10,6	13,6	16	17,4	20	21
50W. - 50A.	1,6	2	2,7	3,2	5	6,8	10,6	12,8	15	16,4	20	21
Terit./Cot.	0,4	0,7	1	1,3	2	2,3	2,8	3,3	3,7	4,1	4,6	5
56W. - 44C.	0,6	1	1,8	2,5	4	5,5	7,5	9	10	11	13	15
33W. - 67C.	0,4	0,9	1,7	2,5	3,5	4,5	6,5	8	9	9,5	11	12
Visc. Wool	1,5	2,5	3,2	4,5	5,5	7	9,5	12	13	14	16	21

- If, for example, "viscose wool" is being processed, for which I want to get a residual humidity by 7%, a value of 6% shall be set according to the table.

SUITABLE OR RECOMMENDED PRODUCTS FOR THE VARIOUS FIBER TYPES

Cotton, cellulose and Liberman fibers	starches, pva, acrylates, cmc
Wool	pva, cmc, acrylates
Silk	acrylic and vinyl copolymers
Staple viscose rayon	acrylic copolymers, cmc
Continuous fiber viscose rayon	acrylic and vinyl copolymers, pva
Cupro thread	acrylic and vinyl copolymers, pva
Acetate rayon	styrene copolymers, acrylic copolymers, pva
Triacetate rayon	acrylic and vinyl copolymers, pva
Polyamide	polyacrylic acids, polymetacrylic acids, copolymeracrylic acids
Continuous polyester	acrylic and vinyl copolymers, soluble pes
Polyacrylic	acrylic and vinyl copolymers
Polypropylene	acrylic and vinyl copolymers, soluble pes
Polyvinyl	acrylic and vinyl copolymers

DRYING TEMPERATURES IN HOT AIR CHAMBER AND CYLINDERS

TIPO DI FIBRA		°C CILINDRI ASCIUGAMENTO											
		°C CAMERA											
		1	2	3	4	5	6	7	8	9	10	11	12
FILATI DISCONTINUI													
ACRILICO	110	125	125	125	125	125	125	120	120	120	120	120	120
COTONE	150	sempre le stesse temperature:materiali poco o relativamente sensibili											
FIOCCO VISCOSA	140	140	140	140	140	140	140	135	135	135	135	135	135
IUTA	150	sempre le stesse temperature:materiali poco o relativamente sensibili											
LANA	120	120	120	120	120	120	120	115	115	115	115	115	115
LINO	130	130	130	130	130	130	130	125	125	125	125	125	125
POLIAMMIDICO	140	125	125	125	125	125	125	120	120	120	120	120	120
POLIESTERE	140	125	125	125	125	125	125	120	120	120	120	120	120
FILATI MISTI													
COTONE-FIOCCO VISCOSA	140	145	145	145	145	145	145	140	140	140	140	140	140
POLIESTERE - LANA	130	125	125	125	125	125	125	120	120	120	120	120	120
POLIESTERE - COTONE	140	135	135	135	135	135	135	130	130	130	130	130	130
POLIESTERE-FIOCCO VISCOSA	140	135	135	135	135	135	135	130	130	130	130	130	130
FILATI TESTURIZZATI													
POLIAMMIDICO	140	120	120	120	120	120	120	120	120	110	110	90	90
POLIESTERE	140	120	120	120	120	120	120	120	120	100	100	90	90
FILI BAVA CONTINUA													
ACETATO	125	80	80	85	85	95	95	95	95	90	90	80	80
FIBRA DI VETRO	150	120	120	120	120	130	130	130	130	120	120	120	120
POLIAMMIDICO	140	130	130	130	130	130	130	120	120	100	100	90	90
POLIESTERE	140	120	120	120	120	120	120	110	110	100	100	90	90
RAION	130	90	90	95	95	110	110	110	110	105	105	90	90

NOTA: Le temperature vanno adattate in funzione dei rispettivi tipi di materiale e secondo le indicazioni del produttore.

USEFUL TABLES

Dry substance on some widely used sizing products

POLYVYNIL ALCOHOL (type LAMOVIL - CARBOVOL - POLYNOL)	95%
CMC (type CARBOCELL - FINFIX)	92%
CMS (type QUICSOLAN - EMSIZE CMS 60) 88%	
MODIFIED STARCHES - eterized esterized (type KOLLOTEX - EMSIZE)	82%
COMPOUNDS (type LAMCOL 750 - C 190 - CARBOTEX)	92%

Calculation of the dry substance on the sized yarn

$$[(\text{Final weight} - \text{Initial weight}) / \text{initial weight}] * 100 = \text{dry substance \%}$$

$$\text{Load on thread \%} = l / \text{size kg} * \% \text{ RS recipe}$$

$$\text{Elongation} = [(\text{final m} - \text{initial m}) / \text{initial m}] * 100$$

Size recipe preparation

$$\text{Sizing product kg} = [(\text{tot. l} * \text{wished conc.}) / \text{product RS \%}]$$

$$\text{Initial water l} = (\text{tot. l} - \text{product kg}) * 0.85$$

0.85 as 15% of the total volume will be composed of condensate

For example: total l 700 - wished concentration 11% - product RS % 85

$$(700 * 11) / 85 = 90.6 \text{ kg of sizing product}$$

$$(700 - 90) * 0.85 = 518.5 \text{ l initial water}$$

Size concentration check

$$[(\text{Product tot. kg} - \text{product RH \%}) / \text{size final vol.}] * 100 = \text{recipe RS \%}$$

Just to give a few useful indications, according to the experience we gained with our customers, some examples of sizing recipes for different types of fibers and yarns follow here below.

1. Sizing of wool

Ne 12 4000 threads recipe = 10% PVA - 0.3% wax
Ne 16 4500 threads recipe = 8% PVA - 0.3% wax
Ne 18 5000 threads recipe = 10% PVA - 0.3% wax

2. Sizing of viscose and its blends

Ne 18 4050 threads recipe = 5% PVA - 0.3% wax
Ne 12 4200 threads recipe = 4% PVA - 0.2% wax
Ne 24 5120 threads recipe = 3% PVA - 0.2% wax
Ne 10.6 3000 threads recipe = 3% PVA - 0.2% wax
Ne 12 2860 threads recipe = 6% PVA - 0.2% wax
Ne 12 5600 threads recipe = 7% PVA - 0.3% wax

3. Sizing of cotton

- Ne 20/1 6430 threads Single size box

PVA	35 kg	
Liq. resin (RS % 25)	1301 t	refractometer = 8.2% TF 4 mm = 10 "
grease	3 kg	

final vol. 850 liters

- Ne 45 7500 threads double size box
- Ne 60 8500 threads double size box

PVA	20 kg	
cmc	10 kg	refractometer = 6
anti-static wax	2 kg	
grease	0.2 kg	

final vol. 500 liters

Ne 40 4570 threads	single size box	
PVA	30 kg	
cmc	40 kg	refractometer = 10.5
grease	5 kg	

final vol. 600 liters

4. Sizing of thread dyed cotton (e.g. denim)

- Ne 6.8 OE continuous line with 1 size box, double squeezing
- | | | |
|-----------------|-------|-------------------|
| Modified starch | 75 kg | |
| Wax | 15 kg | refractometer = 9 |
| Hardener | 25 kg | |

final vol. 1000 l

5. Sizing of a blended cotton yarn

- Ne 30 4500 threads pes / cotton 65 / 35 recipe PVA 8% wax 0.3
- Ne 45 8378 threads pes / cotton 65 / 35

Starch	25 kg	
PVA	50 kg	
CMS	5 kg	
Grease	3 kg	refractometer = 9.5%
Soft	10 kg	

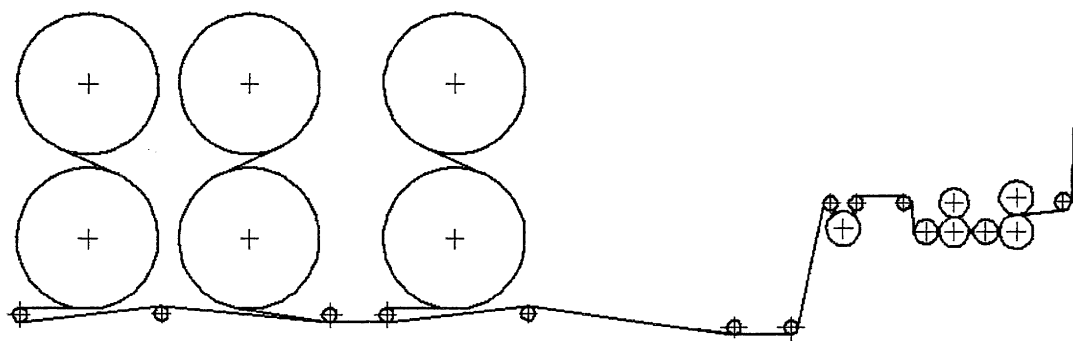
final vol. 830 liters

Sizing recipe for Ne 16 / 1 100%co ring 3556 threads no. 6 beams flange 800*2000
Item = sponge terry cloth

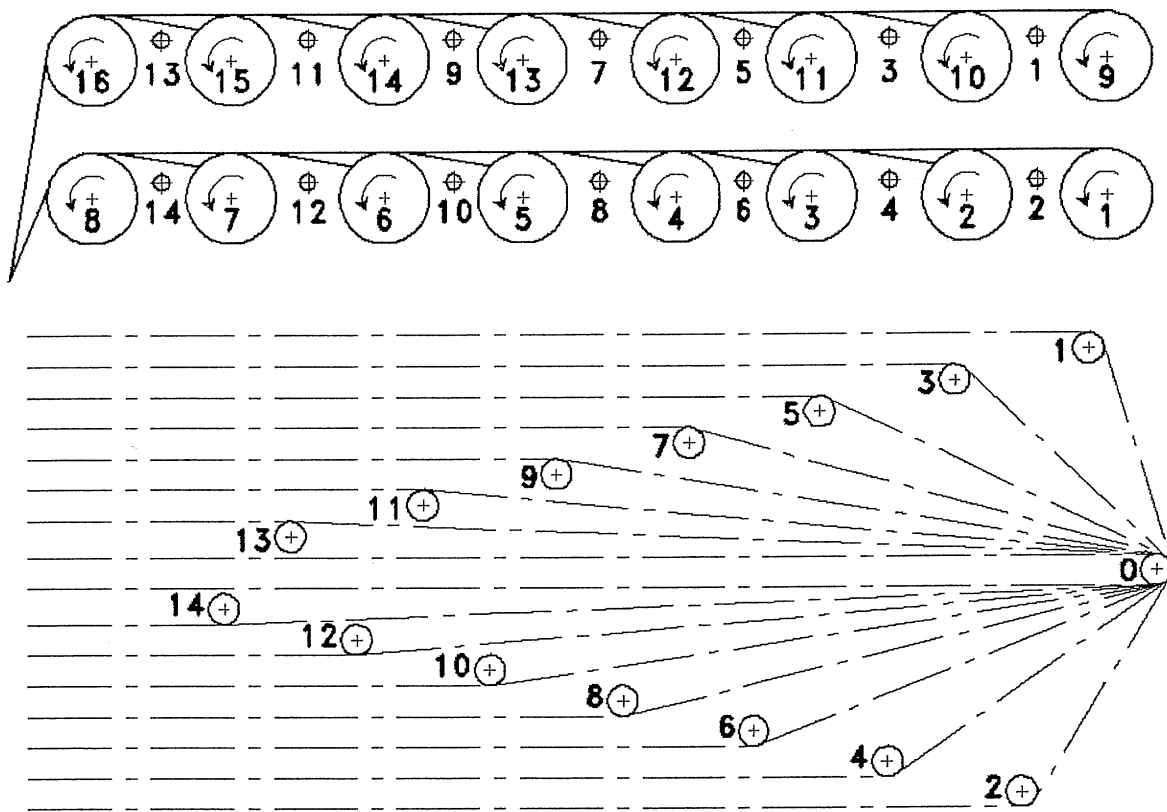
Eterized starch 10 kg	first squeezing 800/1000 KN
Carboxymethylate starch 2 kg	second squeezing 1800/2500 KN
Antistatic wax 2 kg	
Grease 1kg	speed 50 m/min
Water 500 l	RH % = 6.5

Tensions:	classifying	175 KN
	Winding	275 KN
	Carriage pressure	300 KN

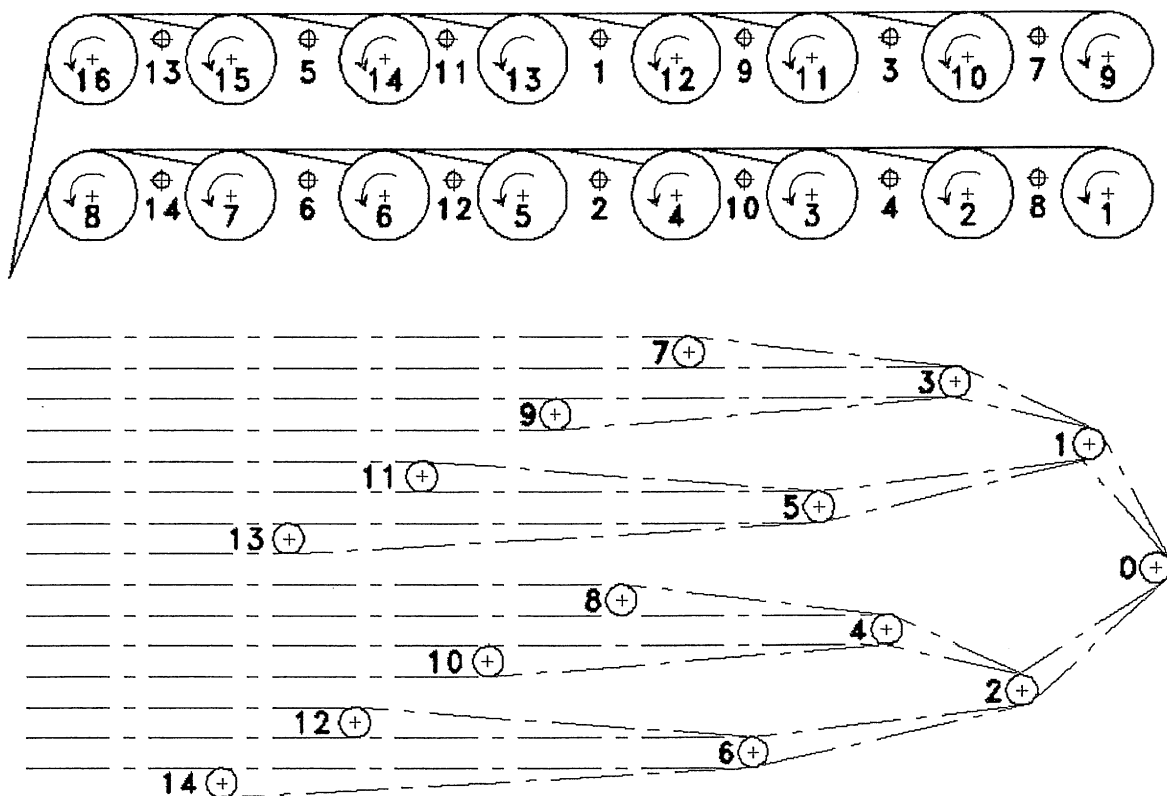
DRAWING-IN



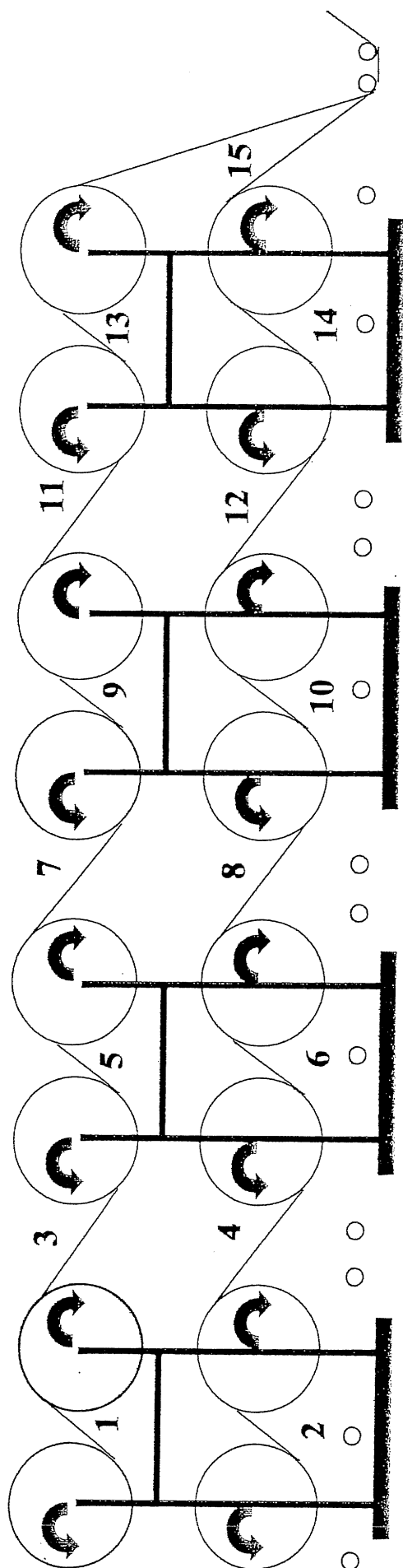
SCHEME 1



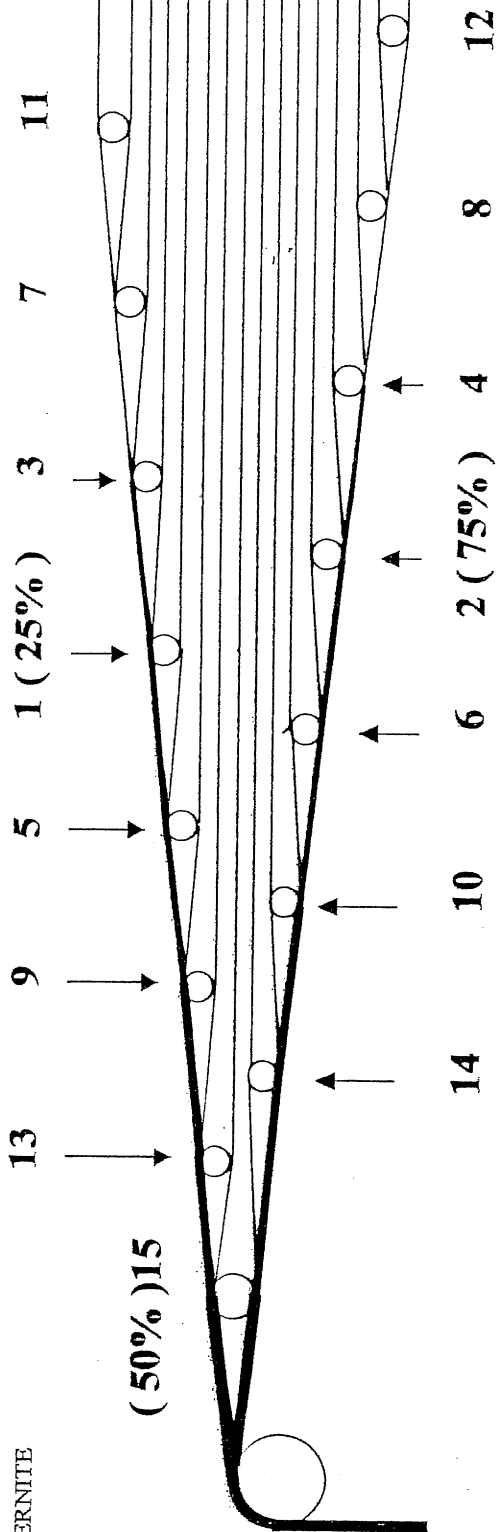
SCHEME 2



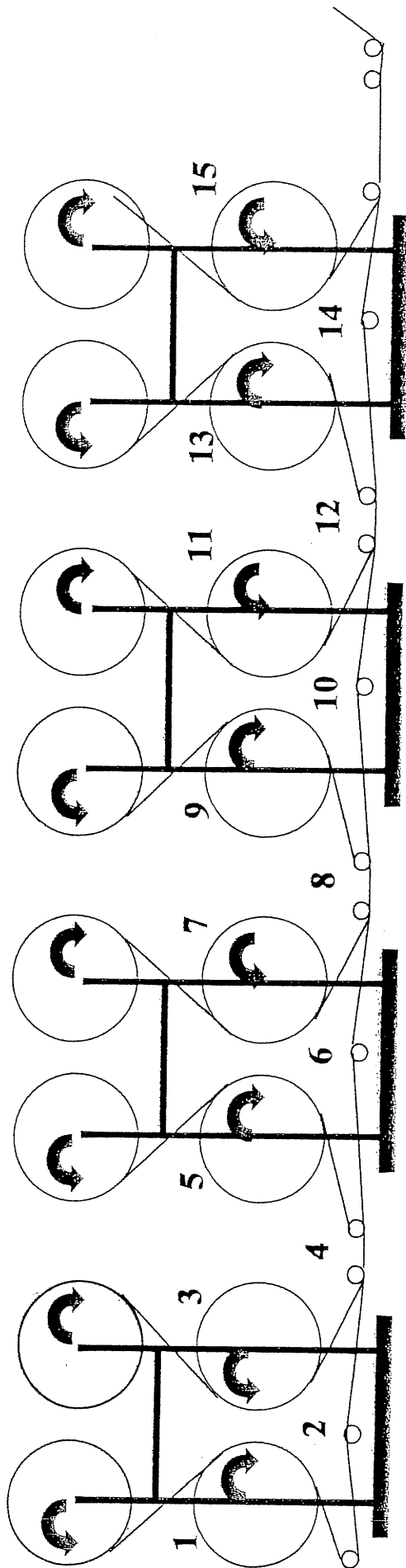
INCORSATURA CANTRA



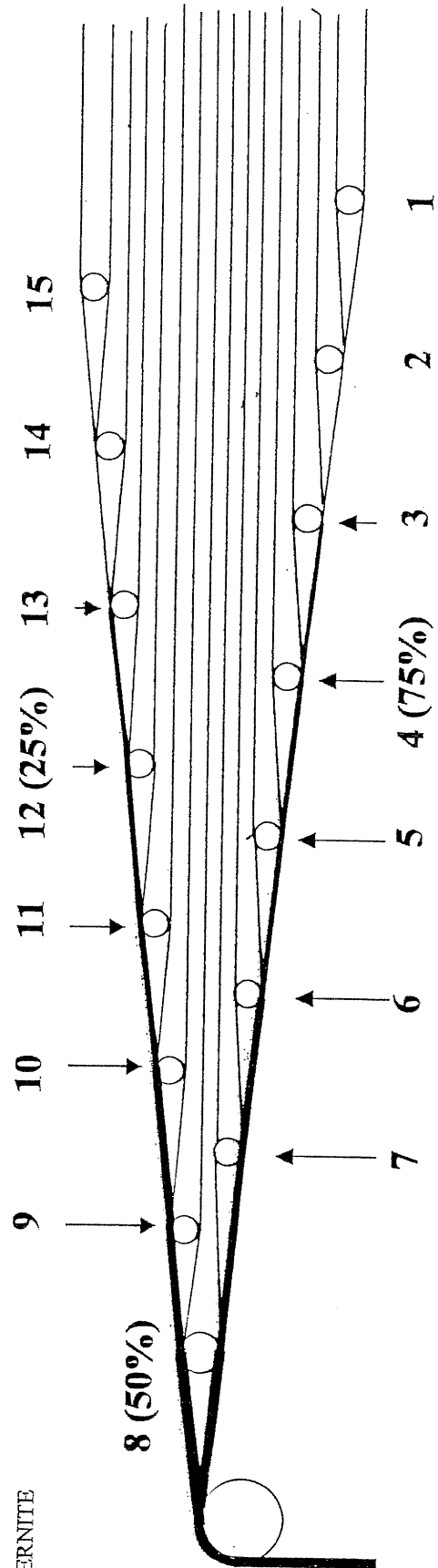
BANCO CERNITE



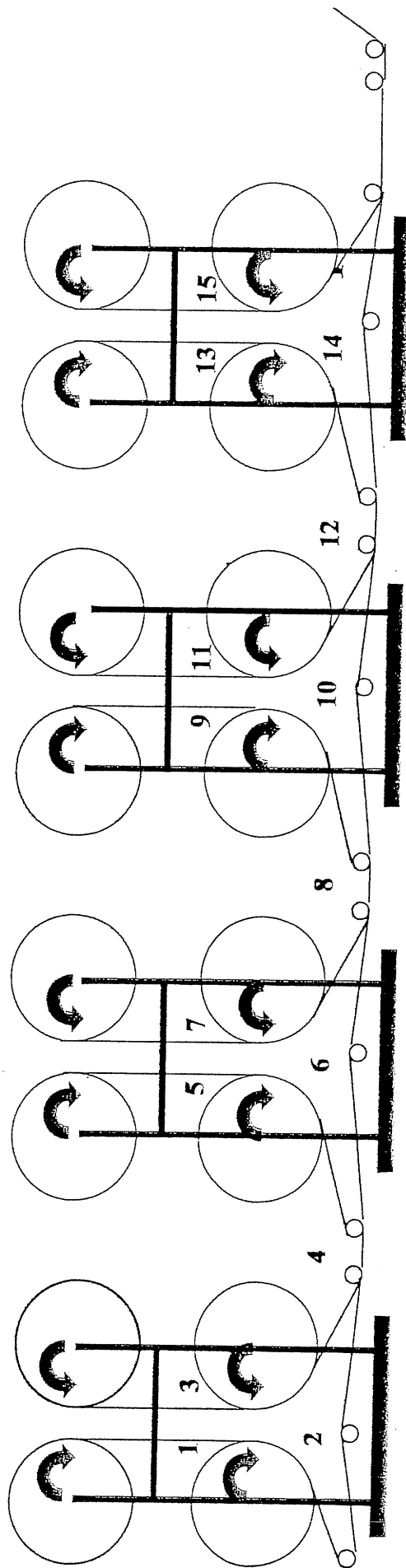
INCORSATURA CANTRA



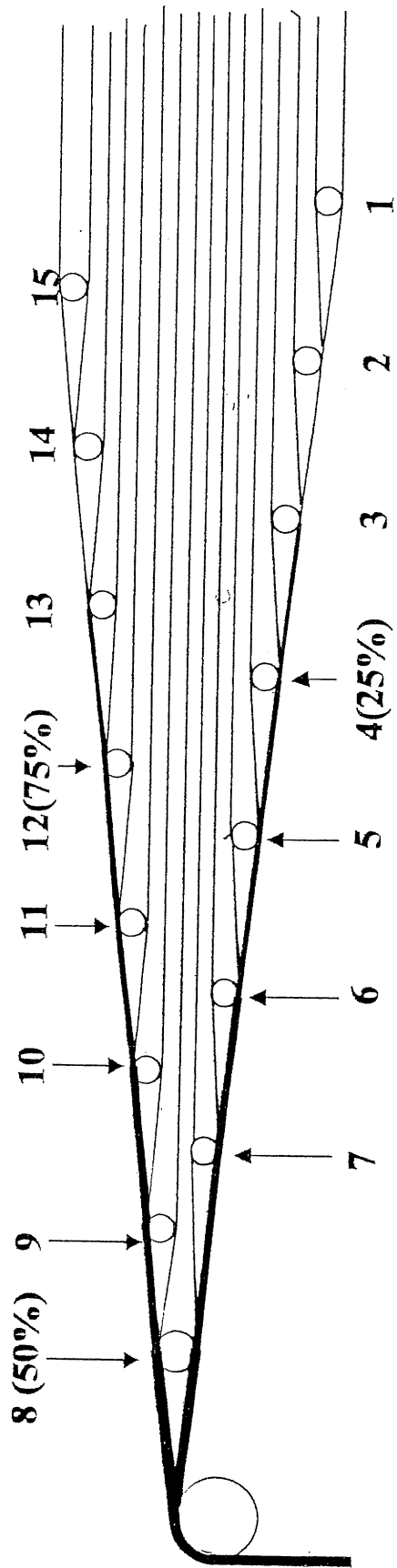
BANCO CERNITE



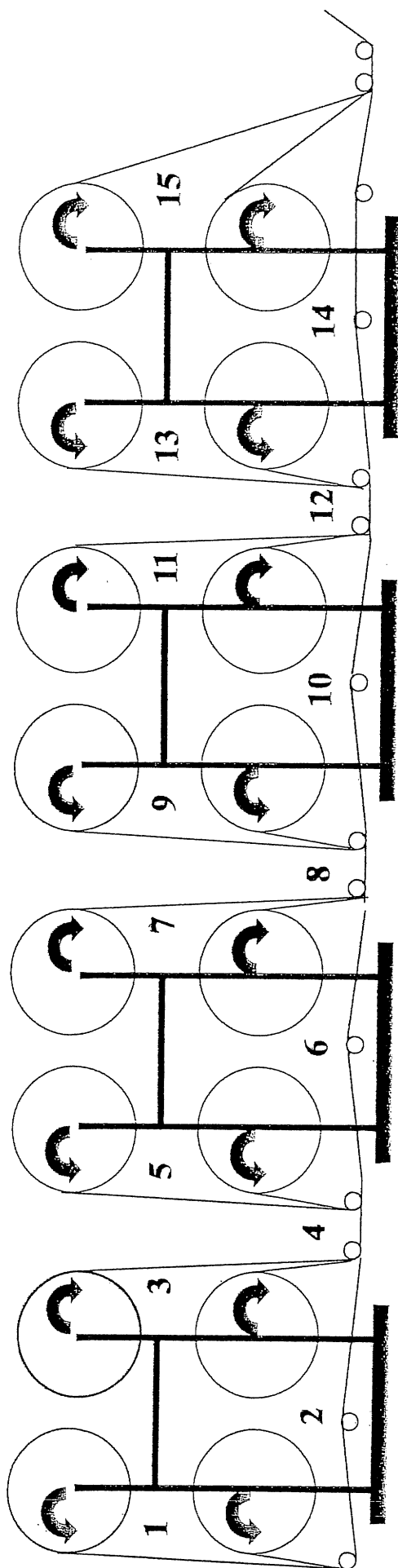
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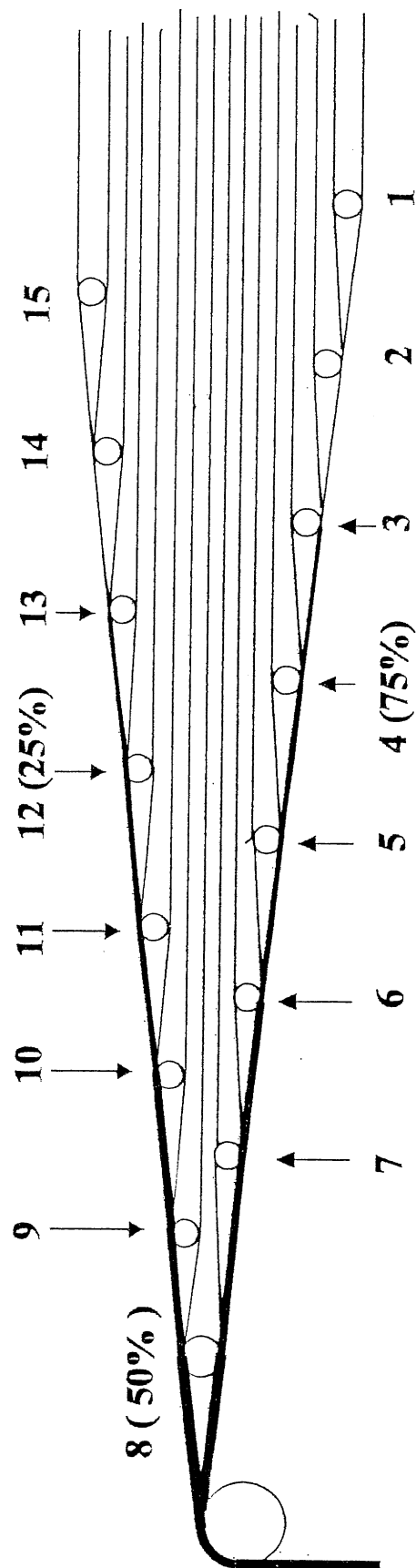
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INCORSATURA CANTRA



BANCO CERNITE



PRODUCT NAME	CATEGORY	SUPPLIER	VISCOSITY	
nome prodotto	categoria	fornitore	viscosità	%rs
quellax 130/130 L	amido esterificato	cognis	bassa	
kollotex 5	amido esterificato	avebe	bassa	81
kollotex 750	amido esterificato	avebe	media-alta	81
avetex n	amido esterificato	avebe	bassa	88
enziplast a 440	amido esterificato	avebe	alta	88
enziplast a 550	amido esterificato	avebe	media-alta	88
emsize e 4	amido esterificato	emsland	bassa	88
emsize e 5	amido esterificato	emsland	alta	88
emsize e 10	amido esterificato	emsland	alta	88
emsize e 19	amido esterificato	emsland	bassa	80
emsize e 21	amido esterificato	emsland	bassa	80
emsize e 6	der amido esterif	emsland	bassissima	84
kollotex 1250	amido eterificato	avebe	bassa	81
kollotex fine	amido eterificato	avebe	bassa	81
kollotex un	amido eterificato	avebe	bassa	81
solvitose xo	amido eterificato	avebe	alta	81
quellax c 255/lv	cms	cognis		
quicsolan cms-lv	cms	avebe	bassa	89
emsize cms 20	cms	emsland	bassissima	87
emsize cms 60	cms	emsland	bassa	87
emsize cms 100	cms	emsland	media	87
emsize cms 150	cms	emsland	alta	87
solvitose gum-dst	pva parz. Idrolizzato	avebe	b - m - a	95
polyvol 4/88	pva parz. Idrolizzato	polychemicals	bassa	95
polyvol 60	pva parz. Idrolizzato	polychemicals	bassa	95
polyvol 70	pva parz. Idrolizzato	polychemicals	bassa	95
polyvol 90	pva parz. Idrolizzato	polychemicals	bassa	95
polyvol 100	pva parz. Idrolizzato	polychemicals	bassa	95
polyvol 200	pva parz. Idrolizzato	polychemicals	media	95
polyvol 200 ak	pva parz. Idrolizzato	polychemicals	media	95
polyvol 300	pva parz. Idrolizzato	polychemicals	media	95
polyvol 400	pva parz. Idrolizzato	polychemicals	alta	95
polyvol 30/88	pva parz. Idrolizzato	polychemicals	alta	95
polyvol 4/98	pva tot Idrolizzato	polychemicals	bassa	95
polyvol 5/99	pva tot Idrolizzato	polychemicals	bassa	95
polyvol 26/99	pva tot Idrolizzato	polychemicals	media	95
inex 746	pva parz. Idrolizzato	cognis	bassa	95
inex 773	pva parz. Idrolizzato	cognis	media	95
lamovil f 105	pva parz. Idrolizzato	lamberti	bassa	95
lamovil f 170	pva parz. Idrolizzato	lamberti	media	95
lamovil f 470	pva parz. Idrolizzato	lamberti	alta	95
polynol 23/88	pva parz. Idrolizzato	polynol	media	95
polynol 55/88	pva parz. Idrolizzato	polynol	alta	95
sico plysran lts	pva parz. Idrolizzato	pjm pujadas	media	93
vicol vl	resina acrilica solida	polychemicals	bassa	94
avirol kw 66	resina acrilica liquida	h - m - r	media	
avirol kw 79	resina acrilica liquida	h - m - r	bassa	
polywax ol	liq. lubrif. antistatico	polychemicals		
polywax extra	emulsione grasso nat	polychemicals		
polywax pe	scaglie-amm. lubrif.	polychemicals		
polywax pl	scaglie-amm. lubrif.	polychemicals		
polywax sg	scaglie-amm. lubrif.	polychemicals		
ekafin il	lubrificante - cere	ekachem		25
ekawax 424	scaglie -ammorb	ekachem		
ekafin 27	lubrific - ammorb	ekachem		
overwax p 45	liq lubrif antistatico	lamberti		
overwax rf	scaglie lubrif-ammorb	lamberti		

SIZE BOX MPD – PW UTILIZATION GUIDELINE

In the pre – wet sizing technology there have been yarn immersion in warm water: that yarn water absorption is in relation with kind of textile material, kind of spinning, yarn count, type of yarn (raw or dyed and which sort of dye).

It's known that OE yarn (rotor technology) has bigger absorption degree than ring yarn (ring spinning) because a yarn structure follow that more open in OE than ring type.

In the dyeing yarn (OE and ring) the absorption of water is higher than raw yarn for marked hydrophilities that change in relation of colour shade (darker is, more hydrophilities has).

In the sizing machine with pre – wet system it's necessary to considerate a size liquor dilution factor to related of water carried by yarn.

The balance should be obtain after 400 / 600 meters of warp and so it's advice to add about 50 liters of water before machine start up to decrease the concentration near 2%, to guarantee same dry size material on yarn, to first till last meter of warp.

At completion of mentioned before :

- 1) The size material concentration decrease about 20% inside size box during process and so it's necessary to get up 1% or 2% recipe's concentration in comparison traditionally ones
- 2) To use always warm water (70°C – 80°C) with raw yarn
- 3) To use always water maximum 50°C with dyeing yarn
- 4) Don't use pre – wet system with warp made by different colour thread
- 5) To consider that not all textile material's yarn can be sized with pre – wet technology polyester, poliammide and in general synthetic or their blend.
For example 35%-65% polyester-cotton yarn could be ok with pre-wet, but the reverse, 65%-35% polyester-cotton doesn't have a good results at loom.

The pre – wet technology advantages are following:

- 1) Size product saving around 30% - 40% in comparison traditionally sizing process
- 2) The warp sized with pre – wet has little bit better loom efficiency.
- 3) In the division table it's possible to have a softer opening layer (also with high concentration)
less dusting, less fibres detachment.
- 4) Less liters of sizing liquor requested because the absorption is lower
- 5) Softer yarn's hand
- 6) Less quantity of size material in waste water in desizing process

SIZE BOX CSB – PW UTILIZATION GUIDELINE

In sizing process of 1 colour dyed warp (indigo or other dyeing) the water temperature in pre-wetting box shouldn't be more than 50°C.

So to avoid adverse discoloration and unwanted shade variation.

Other higher temperature negative effect is that to favouring yarn water absorption and so decrease yarn size material absorption capacity.

This can compromise loom efficiency and fabric quality.

Normally water absorption in pre-wetting system is between 50% and 60% and more dark is the colour, bigger is water's absorption.

Usually a squeezer pressure values is 5 – 5,5 bar, equal 8500-9000 kg.

The warp's high water absorption to entail a dilution effect on liquor in size box and for that reason it's necessary increase from 15% to 20% a size material concentration in cooking device.

About article with different colour thread in the warp are strongly not recommended use pre-wet system: a different discoloration can give yarn different absorption capacity with relative consequence.

In really a CSB-PW size box was planned to be use also without pre-wet function, as MPD-PW size box: a pre-wet lower squeezer roller serve as delivery roller and doesn't used superior squeezer.

About sizing process it's necessary to considerate the size box's structure.

The first roller has immersion roller function and can be use also as side squeezer; if the lateral squeezer it's not use, it's possible to have more size liquor supply, but the thread not accompanied can give a possibility of little bunch of thread.

The second roller is a squeezer and it's possible to used it from 0 to 1200 kg and the pressure values could be keep same for all articles.

The third roller, with inox steel cover, is commanded from motor and has also a immersion roller function.

In that way to form a sizing wedge between last couple of squeezer that lead the final size absorption.

The squeezer parameters should be proportional of machine speed: it's recommended 600/700 kg in slow speed and 1500 – 1800 kg in 2nd or 3rd speed (from 30 to 60 m / minute)

Warning: in that size box is not right to use sizing recipes with high viscosity products, as for example Lamcol 1500 or 2000 and CMC in high concentration.

RUBBER COATING ROLLERS GRINDING

The rubber coatings of squeezing rollers to be subject wear on friction with thread of warp. For this reason, with the passage of the time, happen an squeezer effect reduction between squeezing rollers, and so to became indispensable to grind regularly the rubber coating.

Grinding property

The grinding must be plain and parallel.

Should be done after 2.000 machine working hours.

Independently of preconcerted time of this interval, the squeezing rollers ought to be rectified if during cleaning appear a rubber alteration.

Especially for squeezing rollers with multi-layer and porous rubber coating shouldn't be cleaning before grinding. The rollers have to be dry, hard and so not desized.

The maximum rubber's thickness to grind out possible is 10 mm on the roller coated diameter.

OPERATION	Vs (m/min)	Vw (m/min)	s (mm ^{1°})	a (mm)
Rough-hewing	1800	10	14	0,5
Normal grinding	1800	12,5	8	0,25
Finishing grinding	1800	25	5	0,1

Vs = rotational velocity of stickle

Vw = rotational velocity of roller

s = advance per each roller rotation

a = depth of grinding

To calculate the number of stickle revolution in according with its diameter :

Rotational velocity 1800 m/min

$$n = \frac{1800 \times 1000}{3,14 \times D} \text{ (min}^{-1}\text{)}$$

If the diameter of stickle is D = 400 mm to result that :

$$n = \frac{1800 \times 1000}{3,14 \times 400} = 1432 \text{ min}^{-1}$$

and the number of stickle revolution chosen is $n = 1450 \text{ min}^{-1}$